QuantaRx

Choosing antibiotic combinations with constraint-preserving QAOA

Challenge: The Quantum Optimization Challenge (IBM Qiskit) **Domain:** Healthcare

Stack: Qiskit 2.6.2 · qiskit-optimization 0.7.0 · Qiskit Aer 0.18.2

Code: github.com/naumanzafarprofessional-creator/quantarx-qaoa

> All drug and pathogen numbers are **synthetic** and only loosely based on the literature. They are not a real antibiogram and nothing here is clinical advice. The project demonstrates an optimization method, not a prescribing tool.

1. Problem statement

Motivation. When a patient arrives with severe sepsis, clinicians must start *empiric* antibiotics before lab results identify the pathogen. Every hour of delay raises mortality, so the regimen is chosen under uncertainty. A good combination of drugs should:

- **cover** the pathogens most likely to be present, according to the local antibiogram (how often each drug works against each pathogen);
- **avoid harmful drug–drug interactions**, such as vancomycin + gentamicin (kidney toxicity) or ciprofloxacin + azithromycin (heart-rhythm risk);
- **avoid two drugs from the same class**, which adds toxicity and resistance pressure for little extra coverage;
- **respect antimicrobial stewardship** by holding reserve agents such as carbapenems back where possible.

Intended users. Hospital antimicrobial-stewardship teams who write empiric-therapy guidelines for a ward or setting (ICU sepsis, community pneumonia). The problem is solved once per setting, not once per patient.

Why it is a real optimization problem. Coverage, redundancy and toxicity are all *pairwise* effects: a drug's value depends on which other drugs are in the regimen. The objective is therefore quadratic, and adding up independent scores fails. Even with 10 drugs, a greedy heuristic picks the wrong regimen (PTZ + LZD + GEN, rank 3) instead of the optimum (VAN + CIP + MTZ).

2. Formulation

Decision variables. $x_d \in \{0,1\}$ for 10 drugs: piperacillin-tazobactam (PTZ), meropenem (MEM), ceftriaxone (CRO), cefepime (FEP), vancomycin (VAN), linezolid (LZD), gentamicin (GEN), ciprofloxacin (CIP), azithromycin (AZM), metronidazole (MTZ). This gives **10 qubits**.

Data. Susceptibility S_{dj} (fraction of isolates of pathogen j killed by drug d), pathogen prevalence w_j for each setting, interaction severity T_{de}, same-class indicator D_{de}, stewardship cost c_d.

Coverage. True weighted coverage is non-linear: $\Sigma_j w_j[1 - \Pi_d(1 - S_{dj})^{x_d}]$. Truncating inclusion–exclusion at second order gives $\text{cov}_j(x) \approx \Sigma_d S_{dj}x_d - \Sigma_{d<e} S_{dj}S_{ej}x_dx_e$. The quadratic term is exactly the *redundancy* between two drugs that cover the same pathogen, and the approximation is a provable lower bound (Bonferroni inequality, checked in the tests).

$$\text{minimise}\quad f(x) = -\alpha \sum_j w_j[\sum_d S_{dj}x_d - \sum_{d<e} S_{dj}S_{ej}x_dx_e] + \beta \sum_{d<e} T_{de}x_dx_e + \delta \sum_{d<e} D_{de}x_dx_e + \gamma \sum_d c_dx_d$$

with $\alpha = 1.5$, $\beta = 0.4$, $\delta = 0.3$, $\gamma = 1$

Constraint	Meaning	How it is handled
$\sum_d x_d = 3$	exactly three drugs	penalty $P(\sum x - 3)^2$ in standard QAOA; **enforced by the circuit itself** in XY-QAOA
$x_{VAN} + x_{GEN} \le 1$	at most one kidney-toxic drug	slack-free penalty $P \cdot x_{VAN}x_{GEN}$ (no extra qubits)

Test instance. 10 drugs × 8 pathogen groups, two prevalence settings (ICU sepsis; community-acquired pneumonia). There are 2^{10} = 1,024 assignments, of which 112 are feasible, so every quantum result can be checked against the exact answer.

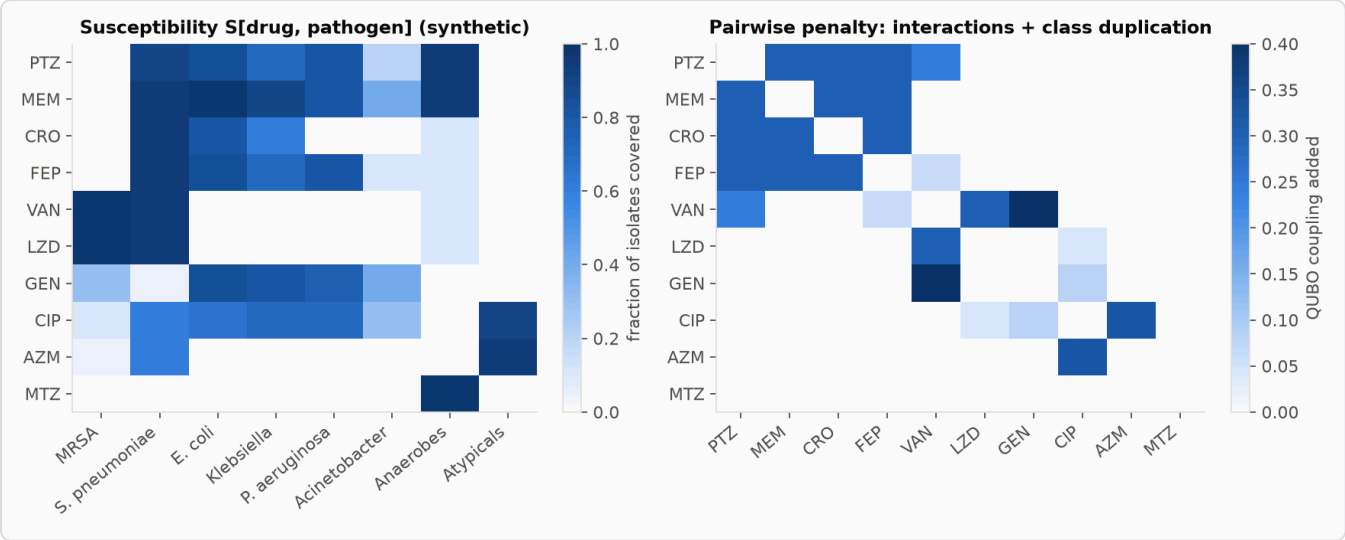


Figure 1. Synthetic instance: drug–pathogen susceptibility (left) and the pairwise penalties from interactions and class duplication (right).

3. Architecture and quantum approach

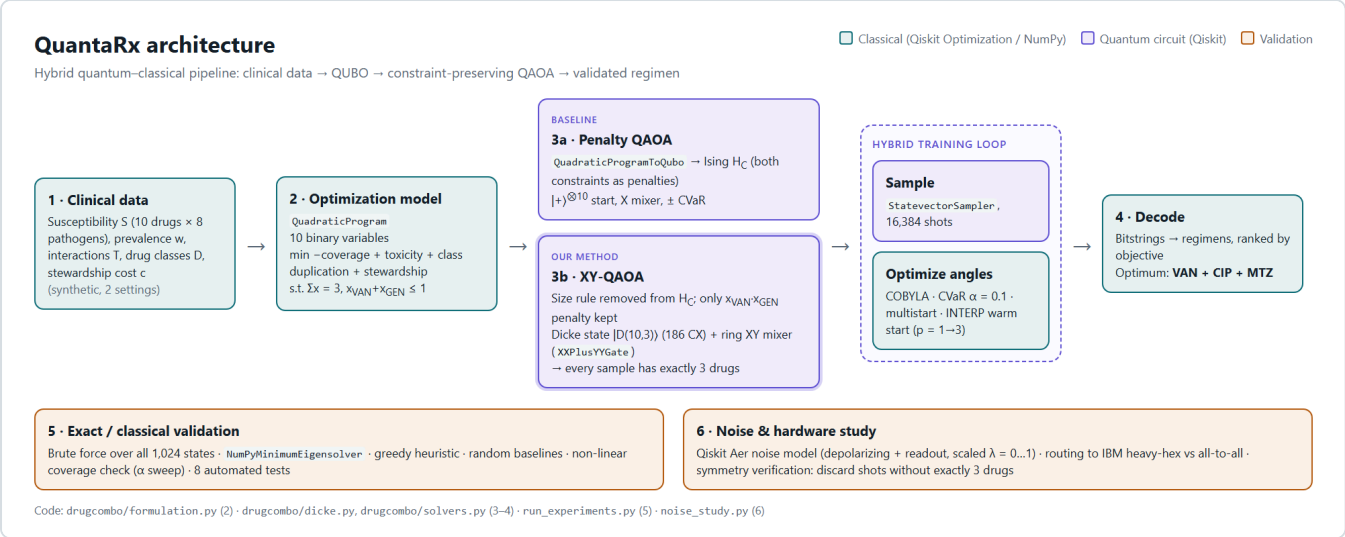


Figure 2. Architecture. Classical modelling (teal) feeds two quantum circuits (violet): the penalty-QAOA baseline and our constraint-preserving XY-QAOA. Angles are trained in a hybrid loop and every result is validated against exact references (orange).

Why QAOA. Every interaction in the model is pairwise, so the problem is a native QUBO / Ising Hamiltonian H_C. The cost unitary $e^{-i\gamma H_C}$ needs only RZ and RZZ gates, which makes QAOA the natural gate-model algorithm.

Our extension: constraint-preserving XY-QAOA. Standard QAOA searches all 1,024 bitstrings and relies on a large penalty to enforce "exactly three drugs". That penalty dominates the energy landscape and hides the clinically meaningful differences between regimens. Instead we:

1. **Prepare a Dicke state** $|D(10,3)\rangle$, an equal superposition of all 120 three-drug regimens. We implemented the Bärtschi–Eidenbenz construction with O(nk) gates: 186 CX, against 1,013 for generic state preparation.
2. **Mix with a ring XY mixer** (`XXPlusYYGate` on neighbouring qubits). Each gate swaps one drug for another and never changes how many are selected.

The circuit therefore never leaves the feasible subspace (leakage $< 10^{-9}$ in tests). Only the small kidney-toxicity penalty remains in H_C.

Training. CVaR aggregation ($\alpha = 0.1$) optimises the best 10% of shots rather than the mean; INTERP warm starts initialise depth p+1 from the optimised depth-p angles; several random starts per depth; COBYLA as the classical optimiser; Hamiltonian normalised by $\sqrt{\Sigma c^2}$ so the angle scale is instance-independent. Sampling uses the Qiskit 2 `StatevectorSampler` (16,384 shots) on a pre-transpiled ansatz.

4. Results and classical / exact comparison

4.1 Exact and classical references (ICU sepsis)

Method	Regimen	Objective
Brute force (all 1,024 states)	**VAN + CIP + MTZ**	**−0.9228**
`NumPyMinimumEigensolver` on the penalised QUBO	VAN + CIP + MTZ	same (penalty is valid)
Optimum of the exact non-linear coverage objective	VAN + CIP + MTZ	QUBO picks the right answer
Greedy heuristic	PTZ + LZD + GEN	−0.8899 (rank 3)

The model makes clinically sensible trade-offs on its own: none of the top-10 regimens pairs vancomycin with piperacillin-tazobactam or combines two β-lactams, and the reserve agent meropenem first appears at rank 9.

4.2 Quantum benchmark (depth p = 3, mean of 3 seeds)

64%
valid regimens, penalty QAOA

99.5%
valid regimens, XY-QAOA

3.5%
P(optimal), XY-CVaR-QAOA

6×
more often optimal than penalty QAOA

Method	P(optimal)	P(top-5)	P(feasible)	Approx. ratio*	CX
QAOA (penalty)	0.0058 ± 0.0002	0.029	0.635	0.752	270
CVaR-QAOA (penalty)	0.0048 ± 0.0005	0.024	0.545	0.749	270
XY-QAOA	0.0149 ± 0.0022	0.077	**0.995**	**0.871**	498
XY-CVaR-QAOA	**0.0346 ± 0.0034**	**0.102**	**0.947**	**0.769**	**498**
Random bitstring	0.0010	0.005	0.109	–	–
Random 3-drug regimen	0.0083	0.042	0.933	0.747	–

*Expected normalised quality of the feasible samples (1 = optimal, 0 = worst feasible). Probabilities are exact Born probabilities at the trained angles.

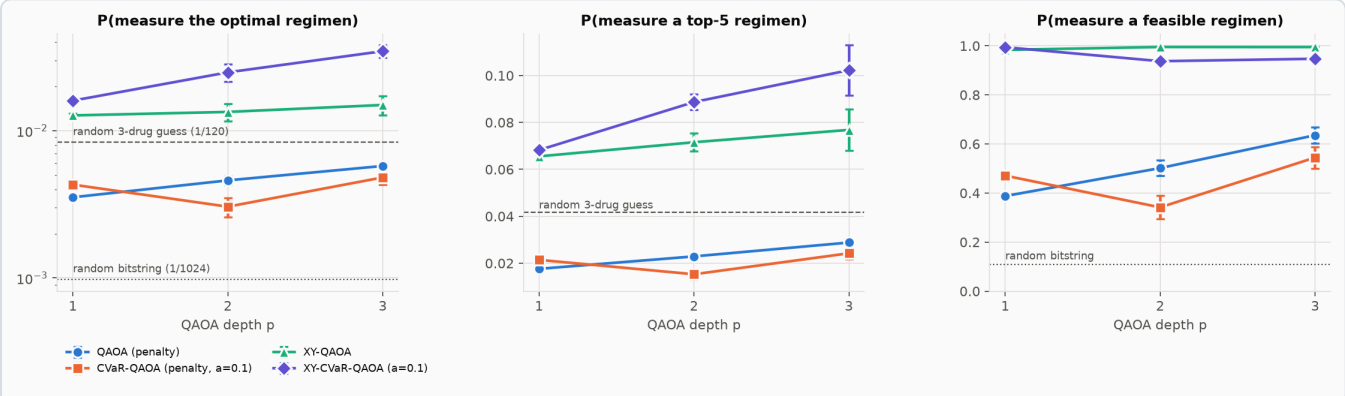


Figure 3. Probability of measuring the optimal regimen, a top-5 regimen and a feasible regimen, against QAOA depth.

Findings.

1. **Penalty QAOA learns feasibility but not quality.** Feasibility rises to 64%, but among feasible samples it is no better than random (approximation ratio 0.752 vs 0.747), and P(optimal) stays below a random three-drug guess. Varying the penalty weight 30× does not fix this.
2. **XY-QAOA fixes the root cause.** Feasibility is 95–99.5% by construction (the remainder is only the kidney-toxicity pair), and P(optimal) grows with depth. XY-CVaR-QAOA finds the optimum **6× more often than penalty QAOA, 4.2× more often than random guessing among three-drug regimens, and 35× more often than random bitstrings.**
3. **The objective matches the goal.** Plain XY-QAOA gives the best average quality (0.871); CVaR concentrates probability on the best regimens (P(top-5) = 0.10), which suits a team reviewing a shortlist.
4. **Second setting.** For community pneumonia the exact optimum is CRO + AZM + MTZ; XY-CVaR-QAOA at p = 3 reaches P(optimal) = 0.085 (10× random three-drug guessing) with 98% feasibility.
5. **QUBO fidelity.** Sweeping α shows the QUBO optimum equals the true non-linear optimum for α ≤ 1.5 (we use 1.5); beyond that the truncation matters.

4.3 Noise and hardware study (Qiskit Aer)

The trained depth-3 circuits were run on a noisy simulator: depolarising gate errors and readout error at roughly current IBM median rates (2-qubit 7.5×10^{-3}, 1-qubit 2.5×10^{-4}, readout 1.5×10^{-2}), scaled by λ from 0 to 1, and routed onto both all-to-all connectivity and an IBM heavy-hex coupling map. Because a correct XY-QAOA circuit always outputs exactly three drugs, any shot with a different count is a **detected error** and is discarded (symmetry verification, at no extra quantum cost). At λ = 0, Aer reproduces the ideal result (0.031 vs 0.030).

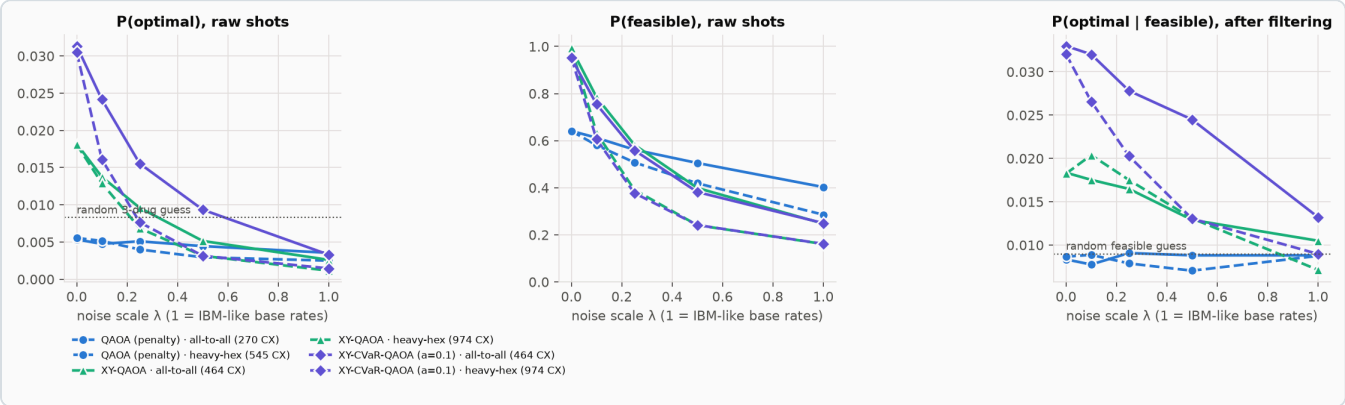


Figure 4. P(optimal) on raw shots, P(feasible), and P(optimal) after discarding shots without exactly three drugs, against noise scale λ (1 = today's rates).

XY-CVaR-QAOA, p = 3	CX	λ = 0	0.1	0.25	0.5	1
all-to-all, after symmetry check	464	0.033	0.032	0.028	0.024	**0.013**
heavy-hex, after symmetry check	974	0.032	0.027	0.020	0.013	0.009
random feasible guess	–			0.0089		

- **Routing roughly doubles the circuit** (464 → 974 CX); connectivity, not qubit count, is the bottleneck.
- **Symmetry verification is a second payoff of the XY design:** at λ = 0.25 on heavy-hex it raises P(optimal) 2.7× while keeping 40% of shots.
- The advantage over random guessing **survives today's noise on all-to-all hardware** (e.g. trapped ions) and up to **about half of today's noise on heavy-hex**. At today's heavy-hex rates it is lost.
- Penalty QAOA carries no signal at any noise level.

5. Limitations and reflection

- **No quantum advantage is claimed.** With 1,024 states, brute force takes milliseconds. The instance is deliberately small so that every quantum result can be verified exactly; the contribution is the modelling and the constraint-aware circuit design.
- **Synthetic data.** The antibiogram, interactions and costs are illustrative, not clinical. Real use would need local antibiogram data and clinical review of the weights α, β, δ, γ.
- **Approximate objective.** The QUBO uses a second-order coverage approximation; it matches the true optimum for $\alpha \leq 1.5$ but not beyond.
- **The most likely bitstring is not always the optimum.** In the best run the optimum is the second most likely outcome. In practice one samples many shots and keeps the best by objective, which is cheap to check classically.
- **Hardware.** On today's IBM heavy-hex devices at depth 3 the method does not beat random guessing; it needs roughly 2× better gates or all-to-all connectivity.
- **Trade-offs observed.** The XY design costs more gates (Dicke preparation, mixer routing) but gives exact feasibility, a cleaner energy landscape and free error detection. CVaR trades average quality for a better shortlist.

6. Reproducibility

All randomness is seeded. The repository README gives setup steps for Windows and Linux/macOS. Main commands: `python -m pytest -q tests` (8 checks, about 30 s), `python run_experiments.py --quick` (demo, about 3 min), `python run_experiments.py` (full benchmark), `python noise_study.py` (noise study). Results and figures are committed in `results/`.